



7 The curve C_1 has polar equation $r = a(\cos \theta + \sin \theta)$ for $-\frac{1}{4}\pi \leq \theta \leq \frac{3}{4}\pi$, where a is a positive constant.

(a) Find a Cartesian equation for C_1 and show that it represents a circle, stating its radius and the Cartesian coordinates of its centre. [4]

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(b) Sketch C_1 and state the greatest distance of a point on C_1 from the pole. [3]

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